

VISHNU TEJA KUNDE

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Skills

Frameworks	Python, PyTorch, GitHub, Hugging Face, Diffusers, Wandb, TRL, Accelerate, PEFT, LoRA
Areas of Expertise	Agentic AI, RL, diffusion LLMs, diffusion/flow models

Selected Publications

Vishnu Teja Kunde*, Fatemeh Doudi*, et al. **Reinforcement Learning for Diffusion LLMs** with Entropy-Guided Step Selection and Stepwise Advantages. *To appear in NeurIPS*, 2026. [Paper] [Code]. *Equal contribution.

Tools/Skills: Diffusion LLMs, RL finetuning, policy gradient techniques, variance reduction.

- Formulated reasoning as multi-step MDP over latent states in masked diffusion LLMs.
- Designed tractable approaches to reduce training compute to take policy gradients on high entropy steps and approximated value with one-step greedy denoised estimate to provided finer credit-assignment.
- Demonstrated state-of-the-art performance on almost all benchmarks.

Vishnu Teja Kunde*, Mahdi Farahbakhsh*, et al. **Inference-Time Search** using Side Information for **Diffusion**-based Image Reconstruction. *To appear in NeurIPS*, 2026. [Paper] [Code]. *Equal contribution.

Tools/Skills: PyTorch, diffusion models, deep learning, probabilistic methods, Bayesian inference.

- Developed a principled, training-free, inference-time search algorithm to improve the performance of diffusion-based reconstruction algorithms for inverse problems using side information.
- Improved the target metric significantly, providing better quality restoration of the ground truth images from noisy measurements.

Professional Experience

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- **Summer Intern**, Qualcomm AI Research, San Diego: developed an agentic harness for multi-hop deep retrieval. May-Aug 2026

Research Projects

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| ◦ LLM for compression : LoRA fine-tuned LLaMA-3.2-1B for text compression, improving rates by 1.5-5%. | 2025 |
| ◦ Transformers & in-context learning : Trained a transformer to predict HMM belief states, matching Baum-Welch. | 2024 |

Education

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| ◦ Ph. D. , Computer Engineering, ECEN, Texas A&M University, Texas, United States | 2022– |
| ◦ M. Tech. , Signal Processing, ECE, Indian Institute of Science (IISc), Bengaluru, India | 2022 |
| ◦ B. Tech. , EEE, National Institute of Technology (NITW), Warangal, India | 2020 |

Relevant Coursework

Reinforcement learning, deep learning, analysis of algorithms, high dimensional probability, advanced convex optimization, matrix theory, random processes, linear and non-linear optimization, measure theory, real analysis, information theory.

Awards and Honors

Masters fellow, CNI, IISc | AIR-179, GATE-EE, 2020 | Merit scholarship, NITW, 2016-20 | AIR-6646, JEE-Mains, 2016